

1. Write a Remote Object Communication Program to create a Student table and display records.

In Eclipse need to add jar file also

In Eclipse need to add jar file also

Query	Query History
1	CREATE TABLE student (
2	student_id INTEGER PRIMARY KEY,
3	student_name VARCHAR(100),
4	student_course VARCHAR(100)
5	);

Data Output	Messages	Notifications
CREATE TABLE		
Query returned successfully in 119 msec.		

Query	Query History
1	INSERT INTO student
2	VALUES
3	(1, 'Prajwal Nevase', 'MCA'),
4	(2, 'Sejal Patole', 'MCA'),
5	(3, 'Harsh Patil', 'MCA'),
6	(4, 'Suyash Tambade', 'MCA'),
7	(5, 'Manish Patil', 'MCA');

Data Output	Messages	Notifications
INSERT 0 5		
Query returned successfully in 73 msec.		

Query

Query History

1SELECT * FROM student;

Data Output

Messages

Notifications

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SQL

	student_id [PK] integer	student_name character varying (100)	student_course character varying (100)
1	1	Prajwal Nevase	MCA
2	2	Sejal Patole	MCA
3	3	Harsh Patil	MCA
4	4	Suyash Tambade	MCA
5	5	Manish Patil	MCA

package

```
RMI_JDBC;
import java.io.Serializable;
public class Student implements Serializable {
    private int student_id;
    private String
student_name; private
String student_course;
    public int getStudent_id() {
        return student_id;
    }
    public void setStudent_id(int student_id) {
        this.student_id = student_id;
    }
    public String getStudent_name() {
        return student_name;
    }
    public void setStudent_name(String
student_name) {
        student_name;
    }
        this.student_name =
    public String getStudent_course() {
```

Name : Prajwal Nevase

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```
    return student_course;
}
public void setStudent_course(String student_course) {
    this.student_course = student_course;
}
}
```

Hello.java (Interface)

```
package RMI_JDBC;
import java.rmi.Remote;
import java.rmi.RemoteException;
import java.util.List;
public interface Hello extends Remote {
    List<Student> getStudents() throws RemoteException, Exception;
    void printMsg() throws RemoteException;
}
```

Server.java**package**

```
RMI_JDBC;
import java.rmi.Naming;
import java.rmi.registry.LocateRegistry;
public class Server {
    public static void main(String args[])
    { try{
        LocateRegistry.createRegistry(9000
    ); Hello obj = new ImplExample();
        Naming.rebind( "rmi://localhost:9000/Hello", obj
    );
        System.out.println("RMI Server Started");
    }
    catch(Exception e) {
        e.printStackTrace();
    }
}
```

Client.java**package**

```
RMI_JDBC;
import
java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
import java.util.List;
public class Client {
    public static void main(String args[])
```

```
{try{
    Registry registry = LocateRegistry.getRegistry("localhost",
    9000); Hello stub = (Hello) registry.lookup("Hello");
    List<Student> list = stub.getStudents();
    for (Student s : list) {
        System.out.println("Prajwal Nevase B2-36");
        System.out.println("Student ID: " + s.getStudent_id());
    }
}
```

```
        System.out.println("Student Name : "+
            s.getStudent_name()); System.out.println("Course : "+
            s.getStudent_course()); System.out.println(" ");
    }
} catch (Exception e) {
    System.out.println("Client Exception : " + e);
    e.printStackTrace();
}
}
```

ImplExample.java

package

RMI_JDBC;

import java.rmi.RemoteException;

import java.rmi.server.UnicastRemoteObject;

import java.sql.Connection;

import

java.sql.DriverManager;

import java.sql.ResultSet;

import java.sql.Statement;

import java.util.ArrayList;

import java.util.List;

public class ImplExample **extends** UnicastRemoteObject **implements** Hello {

protected ImplExample() **throws** RemoteException {

super();

}

@Override

public List<Student> getStudents() **throws** Exception {

List<Student> list = **new** ArrayList<>();

Class.forName("org.postgresql.Driver");

String DB_URL = "jdbc:postgresql://localhost:5432/library";

String USER = "postgres";

String PASS = "studentAditi";

Connection conn = DriverManager.getConnection(DB_URL, USER, PASS);

System.out.println("Database Connected Successfully");

Statement stmt =

conn.createStatement(); String sql =

"SELECT * FROM student"; ResultSet rs

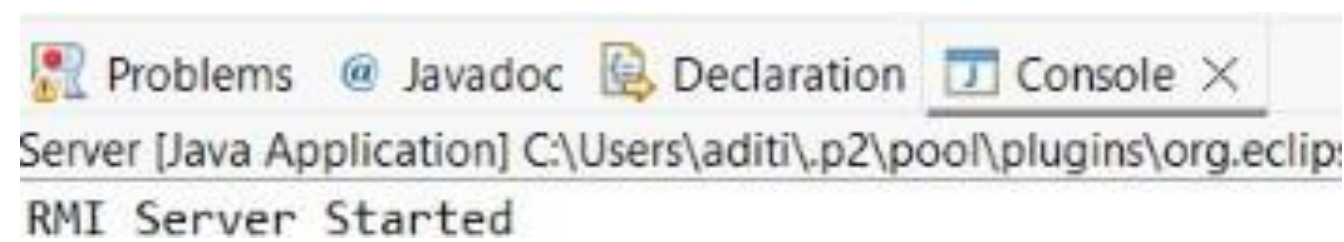
= stmt.executeQuery(sql); **while**

(rs.next()) {

```
Student s = new Student();  
s.setStudent_id(rs.getInt("student_id"));  
s.setStudent_name(rs.getString("student_name"));  
s.setStudent_course(rs.getString("student_course"));  
list.add(s);  
}
```

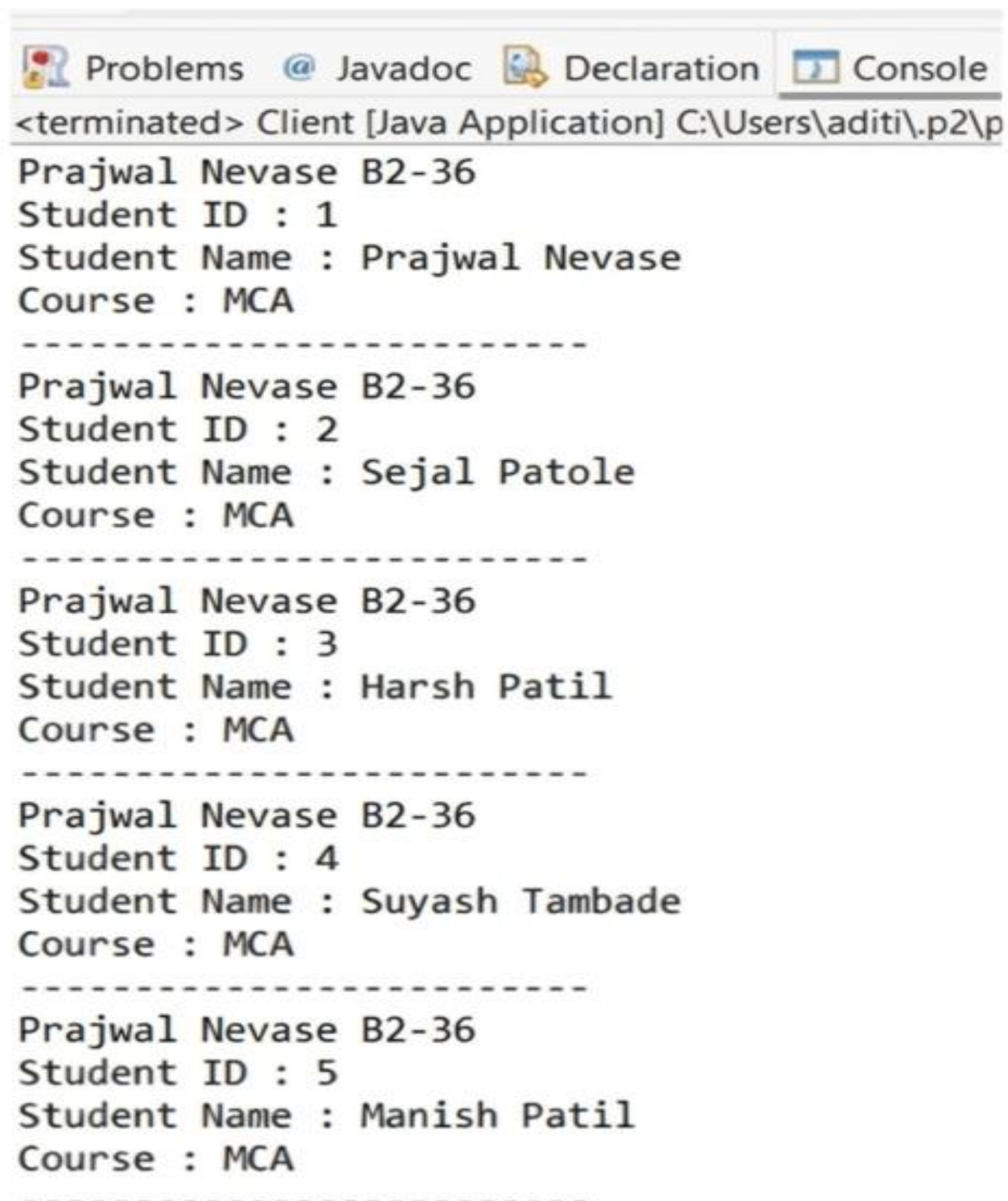
```
rs.close();
stmt.close();
conn.close();
return list;
}
@Override
public void printMsg() throws RemoteException {
    System.out.println("Student RMI Service Working");
}
}
```

Output



Problems Javadoc Declaration Console X
Server [Java Application] C:\Users\aditi\p2\pool\plugins\org.eclipse

RMI Server Started



Problems Javadoc Declaration Console X
<terminated> Client [Java Application] C:\Users\aditi\p2\p

Prajwal Nevase B2-36
Student ID : 1
Student Name : Prajwal Nevase
Course : MCA

Prajwal Nevase B2-36
Student ID : 2
Student Name : Sejal Patole
Course : MCA

Prajwal Nevase B2-36
Student ID : 3
Student Name : Harsh Patil
Course : MCA

Prajwal Nevase B2-36
Student ID : 4
Student Name : Suyash Tambade
Course : MCA

Prajwal Nevase B2-36
Student ID : 5
Student Name : Manish Patil
Course : MCA

2. Write a Remote Object Communication Program to create a electricity Bill table and display records.

Query

Query History

1

2

3

4

5

6

CREATE TABLE electricity_bill (

bill_id INTEGER PRIMARY KEY,

customer_name VARCHAR(100),

units INTEGER,

amount DECIMAL(10,2)

);

Data Output

Messages

Notifications

CREATE TABLE

Query returned successfully in 92 msec.

Query

Query History

1

2

3

4

5

6

7

INSERT INTO electricity_bill

VALUES

(1, 'Prajwal Nevase', 120, 850.50),

(2, 'Sejal Patole', 250, 1750.75),

(3, 'Manish Patil', 180, 1200.00),

(4, 'Suyash Tambade', 300, 2200.50),

(5, 'Harsh Patil', 300, 2200.50);

Query

Query History

1

SELECT * FROM student;

Data Output

Messages

Notifications

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SQL

	student_id [PK] integer	student_name character varying (100)	student_course character varying (100)
1	1	Prajwal Nevase	MCA
2	2	Sejal Patole	MCA
3	3	Harsh Patil	MCA
4	4	Suyash Tambade	MCA
5	5	Manish Patil	MCA

Electricity.java

```
package RMI_JDBC;
import
java.io.Serializable;
public class ElectricityBill implements Serializable {
    private int bill_id;
    private String customer_name;
    private int units;
    private double
amount; public int
getBill_id() {
    return bill_id;
}
public void setBill_id(int bill_id) {
    this.bill_id = bill_id;
}
public String getCustomer_name() {
    return customer_name;
}
public void setCustomer_name(String customer_name) {
    this.customer_name = customer_name;
}
public int getUnits() {
    return units;
}
public void setUnits(int units) {
    this.units = units;
}
public double getAmount() {
```

Name : Prajwal Nevase

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```
    return amount;  
}  
public void setAmount(double amount) {
```

```
    this.amount = amount;
}
}
```

Hello.java

```
package RMI_JDBC;
import java.rmi.Remote;
import java.rmi.RemoteException;
import java.util.List;
public interface Hello extends Remote {
    List<ElectricityBill> getBills()
        throws RemoteException, Exception;
    void printMsg() throws RemoteException;
}
```

Client.java

```
package RMI_JDBC;
import
java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
import java.util.List;
public class Client {
    public static void main(String args[]) {
        try{
            Registry registry =
                LocateRegistry.getRegistry("localhost",9000); Hello stub =
                (Hello) registry.lookup("Hello"); List<ElectricityBill> list =
                stub.getBills();
            for (ElectricityBill b : list) {
                System.out.println("Prajwal Nevase B2-36");
                System.out.println("Bill ID : "+ b.getBill_id());
                System.out.println("Customer Name : "+
                    b.getCustomer_name()); System.out.println("Units : "+
                    b.getUnits()); System.out.println("Amount : ₹ "+
                    b.getAmount()); System.out.println("- ");
            }
        } catch (Exception e) {
            System.out.println("Client Exception : " + e);
            e.printStackTrace();
        }
    }
}
```

Server.java

```
package RMI_JDBC;
import java.rmi.Naming;
```

Name : Prajwal Nevase

Roll No 36 .

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```
import java.rmi.registry.LocateRegistry;  
public class Server {  
    public static void main(String args[]) {  
        try{  
            LocateRegistry.createRegistry(9000);
```

```
        Hello obj = new ImplExample();
        Naming.rebind("rmi://localhost:9000/Hello", obj);
        System.out.println("RMI Server Started");
    }
    catch (Exception e) {
        e.printStackTrace();
    }
}
}
```

ImplExample.java

```
package RMI_JDBC;
import java.rmi.RemoteException;
import java.rmi.server.UnicastRemoteObject;
import java.sql.Connection;
import
java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.ArrayList;
import java.util.List;
public class ImplExample extends UnicastRemoteObject
    implements Hello {
    protected ImplExample() throws RemoteException {
        super();
    }
    @Override
    public List<ElectricityBill> getBills()throws Exception {
        List<ElectricityBill> list = new ArrayList<>();
        Class.forName("org.postgresql.Driver");
        String DB_URL = "jdbc:postgresql://localhost:5432/library";
        String USER = "postgres";
        String PASS = "studentAditi";
        Connection conn =
        DriverManager.getConnection(DB_URL,USER,PASS);
        System.out.println("Database Connected Successfully");
        Statement stmt
        =conn.createStatement(); String sql =
        "SELECT * FROM electricity_bill"; ResultSet
        rs = stmt.executeQuery(sql);
        while (rs.next()) {
            ElectricityBill b = new ElectricityBill();
            b.setBill_id(rs.getInt("bill_id"));
            b.setCustomer_name(rs.getString("customer_name"
            )); b.setUnits(rs.getInt("units"));
        }
    }
}
```

```
        b.setAmount(rs.getDouble("amount"));
        list.add(b);
    }
    rs.close();
    stmt.close();
    conn.close();
    return list;
```

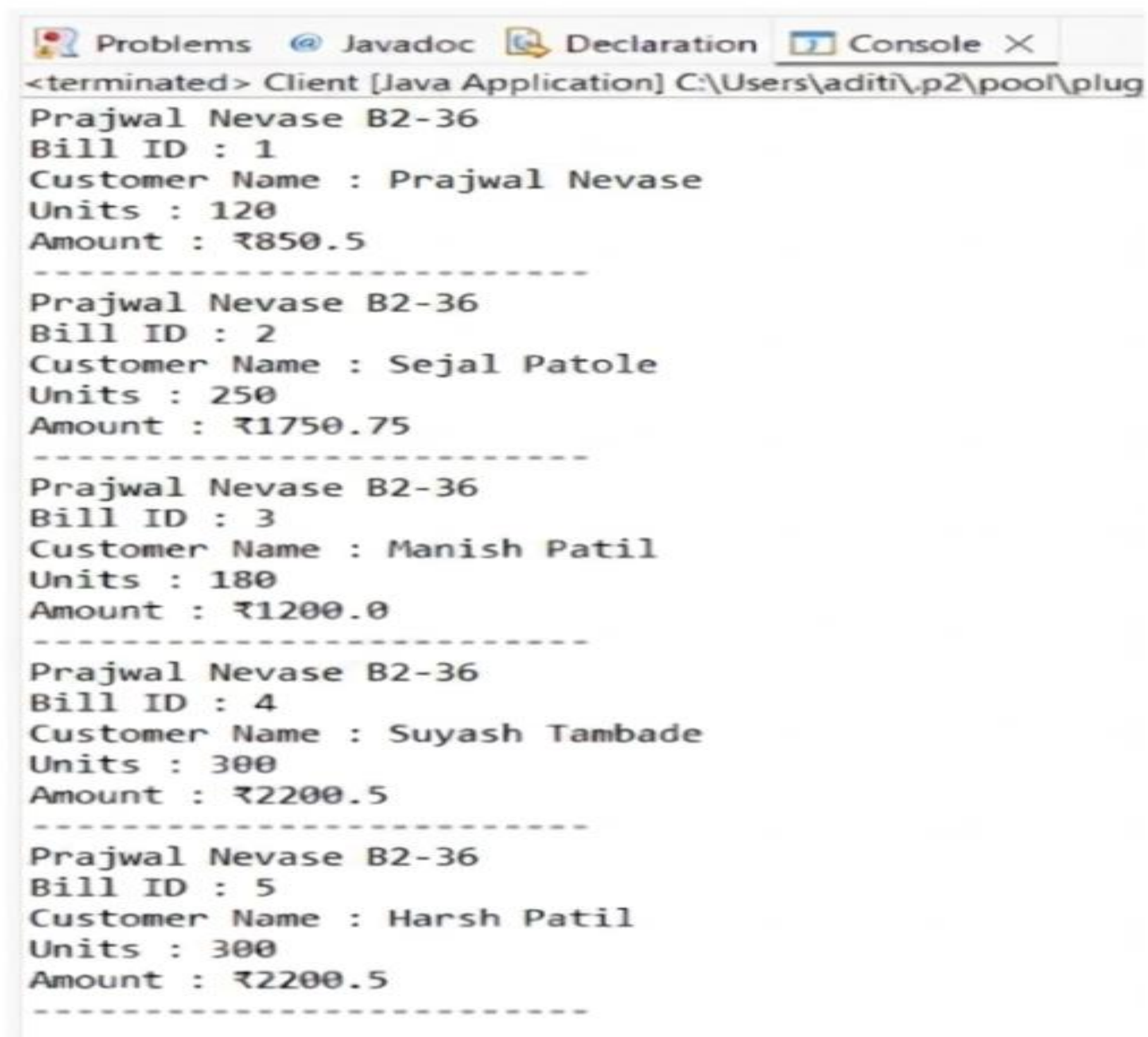
```
}  
@Override  
public void printMsg()  
    throws  
        RemoteException{  
    System.out.println("Electricity Bill RMI Service Working");  
}  
}
```

Output



Problems Javadoc Declaration Console X
Server [Java Application] C:\Users\aditi\p2\pool\plugins\org.eclipse

RMI Server Started



Problems Javadoc Declaration Console X
<terminated> Client [Java Application] C:\Users\aditi\p2\pool\plug

Prajwal Nevase B2-36
Bill ID : 1
Customer Name : Prajwal Nevase
Units : 120
Amount : ₹850.5

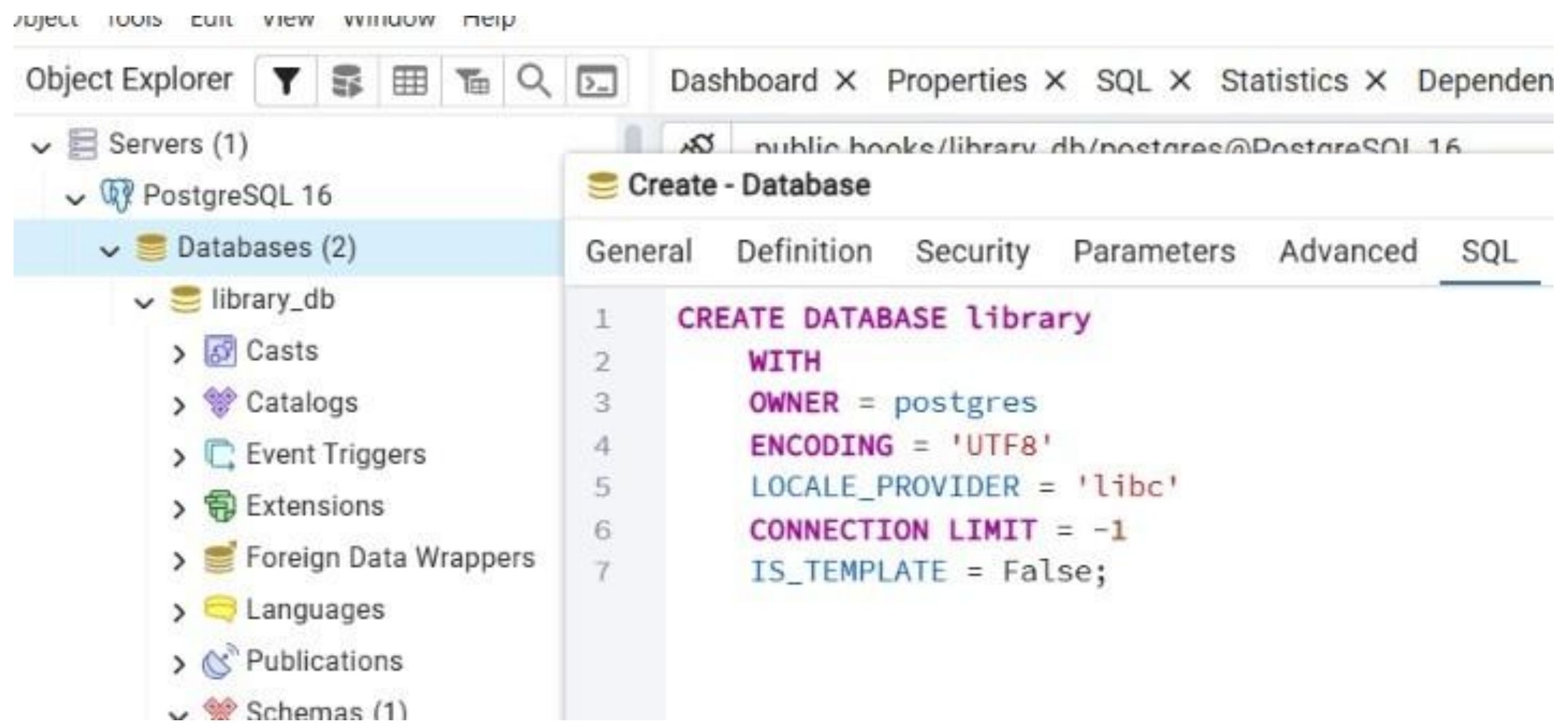
Prajwal Nevase B2-36
Bill ID : 2
Customer Name : Sejal Patole
Units : 250
Amount : ₹1750.75

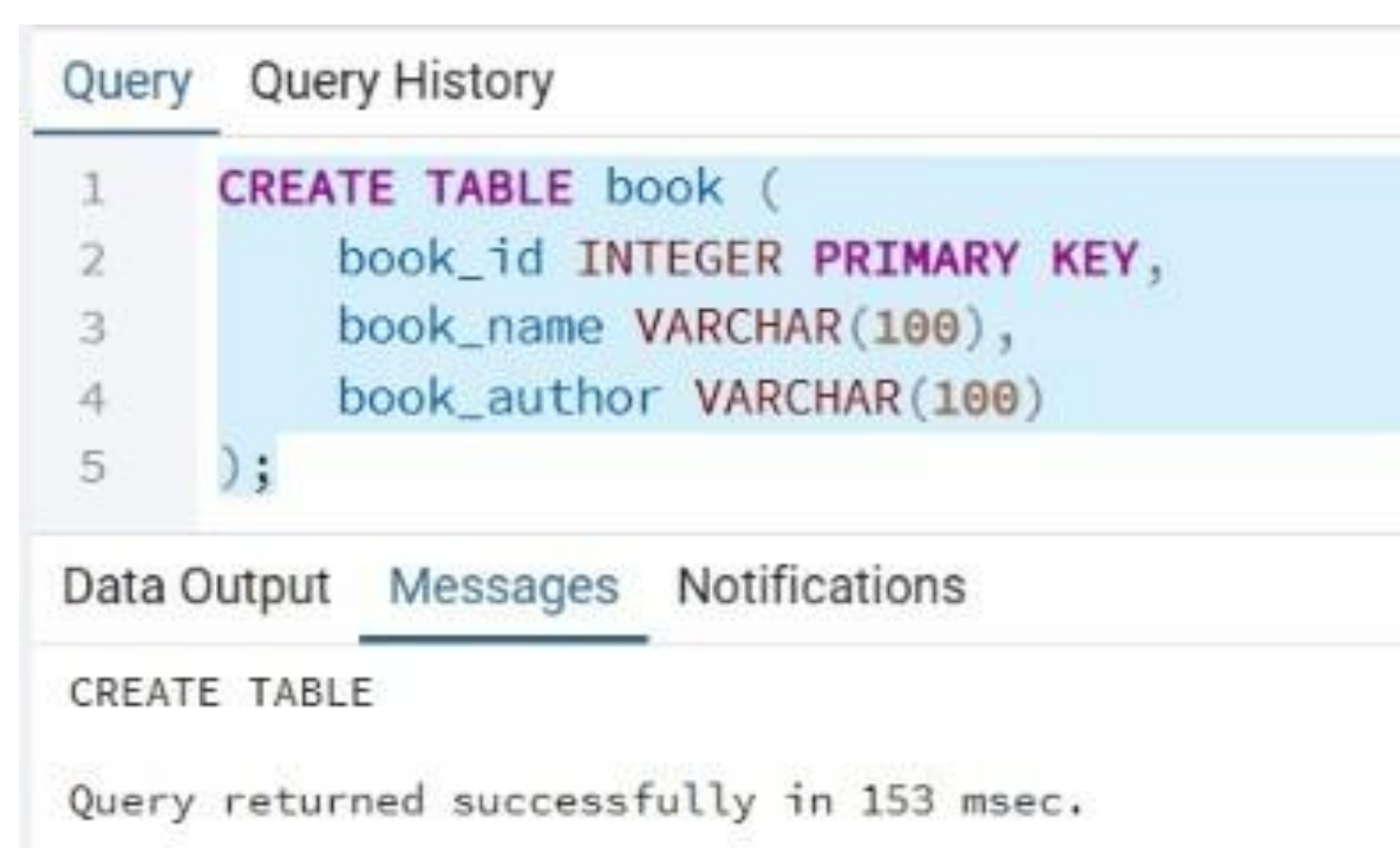
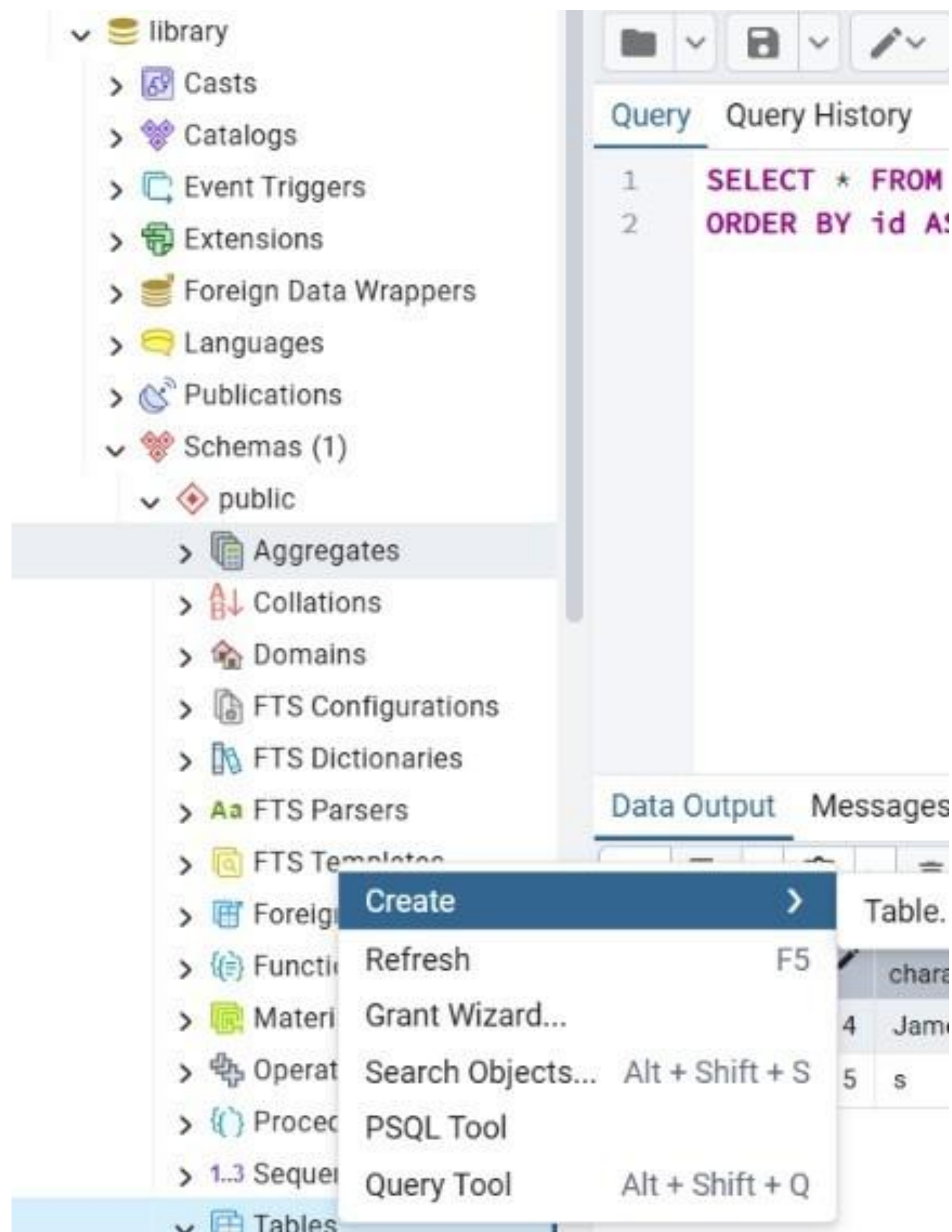
Prajwal Nevase B2-36
Bill ID : 3
Customer Name : Manish Patil
Units : 180
Amount : ₹1200.0

Prajwal Nevase B2-36
Bill ID : 4
Customer Name : Suyash Tambade
Units : 300
Amount : ₹2200.5

Prajwal Nevase B2-36
Bill ID : 5
Customer Name : Harsh Patil
Units : 300
Amount : ₹2200.5

3. Write a Remote Object Communication Program to create a Book table and display records.





Query Query History

1

INSERT INTO book VALUES

2

(1, 'Java Programming', 'Herbert Schildt'),

3

(2, 'Database System Concepts', 'Korth'),

4

(3, 'Computer Networks', 'Andrew Tanenbaum');

Data Output Messages Notifications

INSERT 0 3

Query returned successfully in 142 msec.

Query Query History

1

SELECT * FROM book;

Data Output Messages Notifications

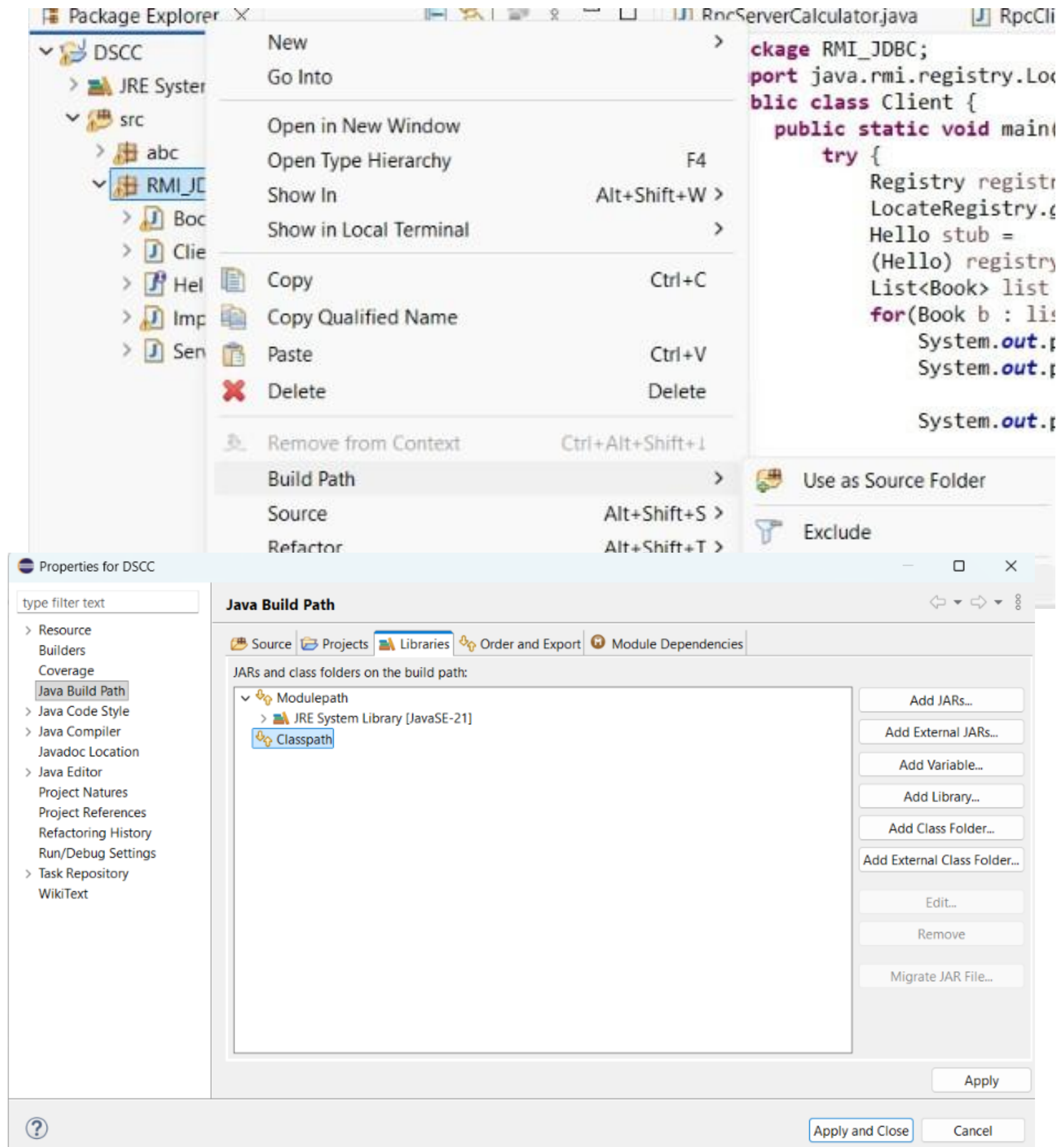
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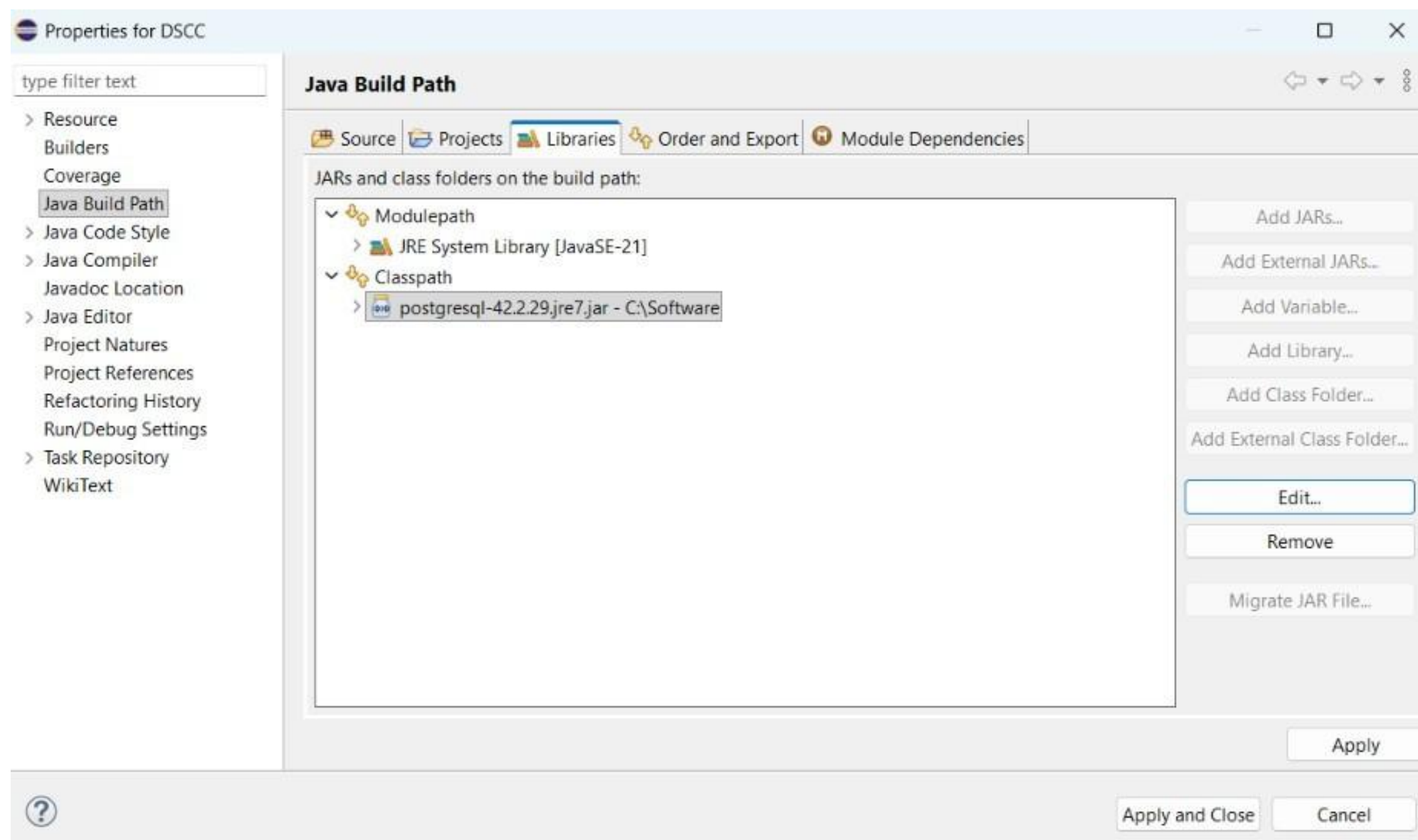
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SQL

	book_id [PK] integer	book_name character varying (100)	book_author character varying (100)
1	1	Java Programming	Herbert Schildt
2	2	Database System Conce...	Korth
3	3	Computer Networks	Andrew Tanenbaum





Book.java

```
packageroc;
import java.io.Serializable;
public class Book implements Serializable{
    private int book_id;
    private String book_name;
    private String book_author;
    public int getBook_id() {
        return book_id;
    }
    public void setBook_id(int book_id) {
        this.book_id = book_id;
    }
    public String getBook_name()
    { return book_name;
    }
    public void setBook_name(String book_name) {
        this.book_name = book_name;
    }
    public String getBook_author() {
        return book_author;
    }
}
```

```
}  
public void setBook_author(String book_author) {  
    this.book_author = book_author;  
}
```

```
}
```

Hello.java —interface

```
packageroc;
```

```
import java.rmi.Remote;  
import  
java.rmi.RemoteException;  
import java.util.List;
```

```
public interface Helloextends Remote {
```

```
    List<Book> getBooks() throws RemoteException, Exception;
```

```
    void printMsg() throws RemoteException;  
}
```

Client.java

```
packageroc;
```

```
import  
java.rmi.registry.LocateRegistry;  
import java.rmi.registry.Registry;  
import java.util.List;
```

```
public class Client {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            Registry registry =  
                LocateRegistry.getRegistry("localhost",9000);
```

```
Hellostub=  
(Hello) registry.lookup("Hello");
```

```
List<Book> list = stub.getBooks();
```

```

for(Book b : list) { System.out.println(
    "Prajwal Nevase B2-36" );
    System.out.println("ID : " + b.getBook_id());

    System.out.println("Name : " + b.getBook_name());

    System.out.println("Author : " + b.getBook_author());

    System.out.println("----- ");
}

}
catch(Exception e) {

    System.out.println("Client Exception : " +

    e);

    e.printStackTrace();
}
}
}

```

Server.java

```

packageroc;
import java.rmi.Naming;
import java.rmi.registry.LocateRegistry;

public class Server {
    public static void main(String args[]) {
        try {
            LocateRegistry.createRegistry(9000);
            Hello obj = new ImplExample();
            Naming.rebind( "rmi://localhost:9000/Hello", obj );
            System.out.println("RMI Server Started");
        }
        catch(Exception e) {

```

```
        e.printStackTrace();
    }
}
```

ImplExample.java

```
packageroc;
import java.rmi.RemoteException;
import
java.rmi.server.UnicastRemoteObject;
import java.sql.Connection;
import
java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.ArrayList;
import java.util.List;

public class ImplExample extends UnicastRemoteObject implements Hello {
    protected ImplExample() throws RemoteException {
        super();
    }
    @Override
    public List<Book> getBooks() throws Exception {
        List<Book> list = new ArrayList<>();

        // PostgreSQL Driver
        Class.forName("org.postgresql.Driver"
        );

        // PostgreSQL Database Details
        String DB_URL = "jdbc:postgresql: //localhost:5432/library";
        String USER = "postgres";
        String PASS = "password";           // Change this to your pgAdmin password

        // Database Connection
        Connection conn = DriverManager.getConnection(DB_URL, USER, PASS);
        System.out.println("Database Connected Successfully by Aditi-B58" );
        Statement stmt = conn.createStatement();

        // Table name created in pgAdmin
        String sql = "SELECT *FROMbook";
        ResultSet rs =
        stmt.executeQuery(sql);
        while(rs.next()) {
```

```
Book b = new Book();
b.setBook_id(rs.getInt("book_id"));
b.setBook_name(rs.getString("book_name"))
;
b.setBook_author(rs.getString("book_author"
)); list.add(b);
}
rs.close();
stmt.close();
```

```
        conn.close();  
        return list;  
    }  
  
    @Override  
    public void printMsg() throws RemoteException {  
        System.out.println("Library RMI Service Working");  
    }  
}
```

Output:

